CONSTRUCTIBILITY IS LOCAL FOR A COVERING BY CONSTRUCTIBLE LOCALLY CLOSED SUBSCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the sole parenthetical about the details of a piece being quasi-compact and quasi-separated concerns a standard property of constructible subsets and the surrounding argument is written out with all its inputs.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Characterization of constructible sheaves by finite partitions. Same chapter, Lemma lemma-constructible-quasi-compact-quasi-separated. Turns constructibility into finite local constancy on the pieces of a partition.
- Constructible sets form a spectral, quasi-compact topology. Topology, Lemma topology-lemma-constructible-hausdorff-quasi-compact together with Properties, Lemma properties-lemma-quasi-compact-quasi-separated-spectral. Lets the author assume the covering is finite.
- Constructibility of a constructible set inside a constructible set, and refinement by a stratification. Topology, Lemmas topology-lemma-constructible-in-constructible and topology-lemma-constructible-partition-refined-by-stratification. Produces the common refinement partition of the whole scheme.

Source reference. Original lemma 0901: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11771–11771. Navigation pointer invoked at proof line 11771; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 094L: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11774–11774. Navigation pointer invoked at proof

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line 11774; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095E: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11717–11717. Navigation pointer invoked at proof line 11780; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09YJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11781–11781. Navigation pointer invoked at proof line 11781; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09Y4: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11745–11745. Navigation pointer invoked at proof line 11745; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $\mathcal{F}$ be a sheaf of sets, abelian groups, Λ-modules (with Λ Noetherian) on $X_{étale}$. If there exist constructible locally closed subschemes $T_i \subset X$ such that (a) $X = \bigcup T_j$ and (b) $\mathcal{F}|_{T_j}$ is constructible, then $\mathcal{F}$ is constructible.*

Proof. First, we can assume the covering is finite as X is quasi-compact in the spectral topology (Topology, Lemma 0901 and Properties, Lemma 094L). Observe that each T_i is a quasi-compact and quasi-separated scheme in its own right (because it is constructible in X; details omitted). Thus we can find a finite partition $T_i = \coprod T_{i,j}$ into locally closed constructible parts of T_i such that $\mathcal{F}|_{T_{i,j}}$ is finite locally constant (Lemma 095E). By Topology, Lemma 09YJ we see that $T_{i,j}$ is a constructible locally closed subscheme of X. Then we can apply Topology, Lemma 09Y4 to $X = \bigcup T_{i,j}$ to find the desired partition of X. □